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MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocols & Architecture
LAB EXPERIMENT # 10

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Score: _____ Signature of the Lab Tutor: _____ Date: _____

Configuring Frame Relay

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	SCORE
OBJECTIVE NO:				

Learning Objectives

1. Configure the link protocol to FR.
2. Configure the working mode of the interface.
3. Configure the VC number of the network.
4. Configure the address map of the interface.

Data Preparation

To connect LANs through the private line, you need the following data:

1. The IP addresses of each interface and the VC number.
2. Router 1 works in DCE mode.
3. Router 2 works in DTE mode.

Frame Relay

Frame Relay is a WAN technology that provides virtual circuits between two devices across a service provider's network. It is a packet-switched WAN protocol, meaning data is sent in small units called frames. Unlike leased lines (point-to-point physical links), Frame Relay allows multiple logical connections over a single physical line.

FR has two types of protocols:

Internet Engineering Task Force (IETF) and nonstandard.

IETF is used to encapsulate the packets according to RFC 1490 specifications and **nonstandard** is used to interconnect with routers of other vendors.

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In FR, both the communicating parties are distinguished as DTE and DCE; they configure the interface in the DTE or DCE format according to their location in the network.

Data Link Connection Identifier (DLCI)

- DLCI is a number assigned to each virtual circuit in a Frame Relay network.
- It is used by Frame Relay switching equipment to identify the path that a frame should take across the service provider's network.
- DLCIs are local identifiers, meaning the same DLCI value can be reused on different physical interfaces without conflict.
- They are included in the Frame Relay frame header, not in the data portion.

Devices that communicate over a serial interface are divided into two classes: **DTE** and **DCE**.

Data Terminal Equipment

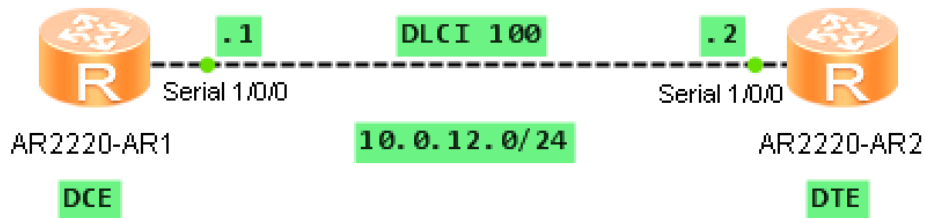
It refers to end devices that create, process, or receive data. In a WAN setup, DTEs are usually customer-side devices like routers, PCs, or switches that connect to the service provider's network. DTE cannot communicate directly over WAN it relies on DCE (Data Circuit-terminating Equipment) to provide the connection and keep the communication path active.

Data Circuit-Terminating Equipment

DCE is a device that helps DTE like a router or computer connect to the service provider's network. It gives the timing (clock) and makes sure the data can travel across WAN. Examples: modem, CSU/DSU, or service provider's router

Topology

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Scenario

Two routers are directly connected through a serial interface. Router 1 works in the FR DCE mode, and Router 2 works in the FR DTE mode.

Tasks

Step 1 Configure Router 1

```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname R1
[R1] interface Serial 1/0/0
[R1-Serial1/0/0] ip address 10.0.12.1 24
[R1-Serial1/0/0] link-protocol fr
Warning: The encapsulation protocol of the link will be changed. Continue? [Y/N]:y
[R1-Serial1/0/0] fr interface-type dce
[R1-Serial1/0/0] fr dlci 100
[R1-fr-dlci-Serial1/0/0-100] q
[R1-Serial1/0/0]
```

Configure dynamic address mapping if the opposite router supports reverse address resolution function.

```
[R1-Serial1/0/0] fr inarp
```

Otherwise, configure static address mapping. The following command maps **local DLCI** with **remote IP address**.

```
[R1-Serial1/0/0] fr map ip 10.0.12.2 100
```

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Step 2 Configure Router 2

```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname R2
[R2] interface Serial 1/0/0
[R2-Serial1/0/0] ip address 10.0.12.2 24
[R2-Serial1/0/0] link-protocol fr
Warning: The encapsulation protocol of the link will be changed. Continue? [Y/N]:y
[R2-Serial1/0/0] fr interface-type dte
[R2-Serial1/0/0] fr dlci 100
[R2-fr-dlci-Serial1/0/0-100] q
[R2-Serial1/0/0]
```

Configure dynamic address mapping if the opposite router supports reverse address resolution function.

```
[R2-Serial1/0/0] fr inarp
```

Otherwise, configure static address mapping. The following command maps **local DLCI** with **remote network layer protocol address**.

```
[R2-Serial1/0/0] fr map ip 10.0.12.1 100
```

Step 3 Verify the configuration

On Router 2, **ping** the interface on **Router 1**.

```
[R2-Serial1/0/0] ping 10.0.12.1
  PING 10.0.12.1: 56 data bytes, press CTRL_C to break
    Reply from 10.0.12.1: bytes=56 Sequence=1 ttl=255 time=40 ms
  Reply from 10.0.12.1: bytes=56 Sequence=2 ttl=255 time=40 ms          Reply
  from 10.0.12.1: bytes=56 Sequence=3 ttl=255 time=30 ms
    Reply from 10.0.12.1: bytes=56 Sequence=4 ttl=255 time=40 ms
  Reply from 10.0.12.1: bytes=56 Sequence=5 ttl=255 time=50 ms

  --- 10.0.12.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
  0.00% packet loss
    round-trip min/avg/max = 30/40/50 ms
```

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Display the FR address mapping on Router 2. In the following example, the FR interface learns the peer DLCI through dynamic address mapping.

```
[R2-Serial1/0/0] display fr map-info interface Serial 1/0/0
Map Statistics for interface Serial1/0/0 (DTE)      DLCI
= 100, IP INARP 10.0.12.1, Serial1/0/0      create
time = 2019/06/03 18:53:10, status = ACTIVE
encapsulation = ietf, vlink = 1, broadcast
```

Run the **display fr inarp-info** command. If statistics about the FR RARP are displayed, it means the configuration succeeds.

```
[R2-Serial1/0/0] display fr inarp-info
Frame relay InverseARP statistics for interface Serial1/0/0 (DTE)
  In ARP request  Out ARP reply  Out ARP request  In ARP reply
  12              12             13              13
```

Final Configuration

```
[R1] display current-configuration
[V200R003C00]
#
sysname R1 #
interface Serial1/0/0
link-protocol fr fr
interface-type dce
fr dlci 100
ip address 10.0.12.1 255.255.255.0
#
Return
```

```
[R2] display current-configuration
[V200R003C00]
# sysname
R2 #
```

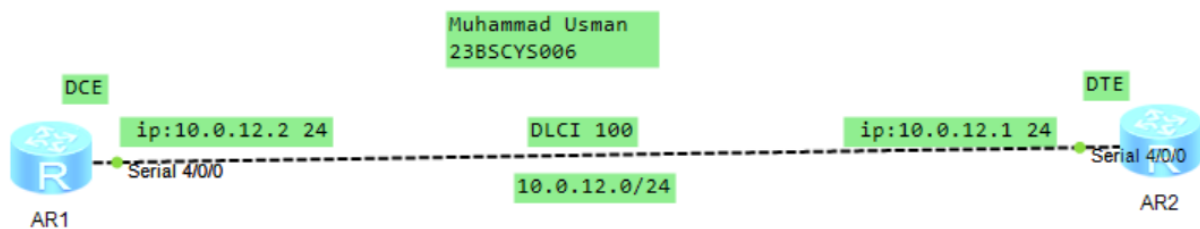
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```
interface Serial1/0/0
link-protocol fr fr
dlci 100
ip address 10.0.12.2 255.255.255.0
# return
```

Tasks

Try a different topology and demonstrate your learning.

Upload the `.topo` `.efz` `.cfg` files



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```
[23BSCYS006-R1]dis current-configuration
[V200R003C00]
#
sysname 23BSCYS006-R1
#
board add 0/4 2SA
#
snmp-agent local-engineid 800007DB03000000000000
snmp-agent
#
clock timezone China-Standard-Time minus 08:00:00
#
portal local-server load flash:/portalpage.zip
#
drop illegal-mac alarm
#
wlan ac-global carrier id other ac id 0
#
set cpu-usage threshold 80 restore 75
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$%$K8m.Nt84DZ)e#<0`8bmE3Uw)%$%$
local-user admin service-type http
#
firewall zone Local
priority 15
#
#
firewall zone Local
priority 15
#
interface Serial4/0/0
link-protocol fr
fr interface-type dce
fr dlci 100
fr map ip 10.0.12.2 100
ip address 10.0.12.1 255.255.255.0
#
interface Serial4/0/1
link-protocol ppp
#
interface GigabitEthernet0/0/0
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
user-interface con 0
authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
[23BSCYS006-R1]
```

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```
[23BSCYS006-R2]dis current-configuration
[V200R003C00]
#
sysname 23BSCYS006-R2
#
board add 0/4 2SA
#
snmp-agent local-engineid 800007DB0300000000000000
snmp-agent
#
clock timezone China-Standard-Time minus 08:00:00
#
portal local-server load flash:/portalpage.zip
#
drop illegal-mac alarm
#
wlan ac-global carrier id other ac id 0
#
set cpu-usage threshold 80 restore 75
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$%$K8m.Nt84DZ)e#<0`8bmE3Uw}%$%$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface Serial4/0/0
link-protocol fr
fr dlci 100
fr map ip 10.0.12.1 100
ip address 10.0.12.2 255.255.255.0
#
interface Serial4/0/1
link-protocol ppp
#
interface GigabitEthernet0/0/0
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
user-interface con 0
authentication-mode password
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
[23BSCYS006-R2] |
```